

Programme	B. Sc. Computer Science				
Course Title	Fundamentals of System Software, Networks and DBMS				
Type of Course	Minor				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic understanding of computer operation Basic Science fundamentals				
Course Summary					
The course covers essential concepts in operating systems, network protocols, and database management systems, providing foundational knowledge for computer science and IT careers.					

Course Outcomes (CO): .

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand System Software principles	U	C	Instructor-created Exams / Assignment/ Viva Voce
CO2	Understand basic concepts of operating systems functions	U	C	Instructor-created Exams / Assignment/ Viva Voce
CO3	Interpret the concepts of data communications system and its components	An	C	Instructor-created Exams / Assignment/ Viva Voce

CO4	Acquire a good understanding of the architecture and functioning of Database Management Systems.	U	C	Instructor-created Exams / Assignment/ Viva Voce
CO5	Construct basic SQL queries to retrieve and manipulate data as required.	C	P	Practical/Exam/ Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I			11	18
	1	Overview of System software and Application Software	1	
	2	System Software Components: operating systems, compilers, and device drivers.	1	
	3	Compilers: Classification of programming languages and language processors	2	
	4	Types of Operating System	3	
	5	Functions of Operating System	3	
II		Computer networks	12	18
	6	Goals of networking	1	
	7	network topologies	1	
	8	types of networks (LAN, MAN and WAN)	1	
	9	Communication Media-Guided (Twisted Pair, Coaxial Cable and Fiber Optic) and	2	
	10	Communication Media -Unguided (microwave, satellite)	2	
	11	Network OSI model- 7 layers	3	

	12	Internet Layer- 5 layers	2	
III	Database Management Systems		12	18
	13	Introduction to DB and DBMS:Definition	1	
	14	Advantages of DBMS	1	
	15	Three schema architecture of DBMS(External,Conceptual and internal)	1	
	16	Data Independence: Logical data independence and Physical data independence	2	
	17	Structure of Database Management System	2	
	18	Data models (Relational Model,Network Model. c. The Hierarchical Model. Object-Oriented Model)	5	
IV	Structured query language - Create, insert, select, update, delete, alter, drop commands		10	16
	19	DML	2	
	20	DDL	3	
	21	Constraints	2	
	22	Operators and functions	3	
V	Practical Applications, Case Study and Course Project		30	
	1	1. Create Database: ● Write a SQL query to create a new database in MySQL. 2. Create Table: ● Create tables with various data types for columns such as INT, VARCHAR, DATE, etc. ● Include constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, etc. 3. Insert Data: ● Insert records into tables using the INSERT INTO statement. ● Practice inserting data into tables with	20	

		different data types. 4. Retrieve Data: ● Write SELECT queries to retrieve data from tables. ● Retrieve specific columns using SELECT. ● Filter rows using the WHERE clause. 5. Update Data: ● Update existing records in a table using the UPDATE statement. ● Modify records based on specific conditions using the WHERE clause. 6. Delete Data: ● Delete records from a table using the DELETE statement. ● Remove records based on specific conditions using the WHERE clause. 7. Sorting and Filtering: ● Sort the result set using ORDER BY clause. ● Filter records using various conditions such as equality, comparison operators, and logical operators. 8. Grouping and Aggregation: ● Group rows using GROUP BY clause. ● Use aggregate functions like COUNT(), SUM(), AVG(), MIN(), and MAX(). 9. String Functions: ● Use string functions like CONCAT(), SUBSTRING(), UPPER(), LOWER(), etc. ● Manipulate string data in SELECT queries. 10. Date and Time Functions: ● Use date and time functions like DATE(),		
--	--	---	--	--

		NOW(), YEAR(), MONTH(), DAY(), etc. ● Work with date and time data in SELECT queries. 11. Mathematical Functions: ● Use mathematical functions like ROUND(), CEIL(), FLOOR(), ABS(), etc. ● Perform mathematical operations on numeric data in SELECT queries. 12. Conditional Functions: ● Use conditional functions like IF(), CASE statement, etc. ● Implement conditional logic in SELECT queries.		
		Case study: 1. Library Management System: Track books, borrowers, and transactions, facilitating library operations efficiently. 2. Student Information System: Manage student records, courses, grades, and attendance for academic institutions. 3. Employee Database System: Store employee details, salaries, and performance evaluations, streamlining HR processes for companies.	10	

References

- 1 P. K Sinha, Fundamentals of Computers
2. D. M Dhamdhere, Operating System: A concept based Approach
3. Behrouz A Forouzan, Data Communication & Networking, MC Graw Hill
4. "Learning MySQL: Get a Handle on Your Data" by Seyed M.M. (Saied) Tahaghoghi and Hugh E. Williams.

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO 1	PO 2	PO 3	PO 4	PO 5	P O 6
CO 1	-	2	1	-	1	3						
CO 2	-	2	1	-	1	3						
CO 3	-	2	1	-	1	3						
CO 4	-	2	1	-	1	3						
CO 5	-	2	1	-	1	3						
CO 6	-	-	2	-	1	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignme nt	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓